

Retrieval-Augmented Generation

Overview

Retrieval-Augmented Generation (RAG) combines information retrieval with a language model. Before generating an answer, the system searches a knowledge collection for relevant passages and provides those passages as context to the language model.

RAG Pipeline

A typical RAG pipeline loads documents, divides them into smaller chunks, converts the chunks into vector embeddings, stores the vectors in a vector database, retrieves relevant chunks for a user question, and supplies the retrieved context to a language model.

Benefits

RAG can help a language application answer questions about private, domain-specific, or frequently changing documents without requiring the language model to contain every piece of information in its original training data. Retrieved passages can also be displayed as sources.

Limitations

RAG quality depends on retrieval quality. Poor chunking, incomplete documents, ambiguous questions, or irrelevant retrieval can lead to weak answers. A language model may also produce unsupported statements if the application does not constrain generation and evaluate the retrieved context.